

Does the Mean-Field Machinery Earn Its Keep?

An ablation of calibration and reputation in a production LLM judge panel

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Abstract

Hanzo ships a Mean-Field Judge Panel (MFJP): a population of diverse LLM judges whose raw scores are per-judge z-score calibrated and then combined with multiplicative-weights reputation. We ask whether the two mean-field mechanisms — calibration and reputation — earn their keep against the obvious alternatives, and we set out to falsify rather than confirm. Two tracks: 164 HumanEval candidates labelled by *executing* their tests, scored by three real judges from different model families under the verbatim production rubric; and a Monte-Carlo study (400 replicates) that drives the *shipped panelScore* through regimes chosen so the panel can lose. The ablation is surgical: identical online calibration with reputation weights pinned to 1 isolates reputation, and a plain mean of raw scores isolates calibration. **The headline is negative, and it holds on real data.** On the real judges, a plain mean of the same three scores significantly outperforms the shipped panel (AUC 0.859 vs 0.824, paired $\Delta = +0.035$, 95% CI $[+0.014, +0.057]$), while the calibration-only ablation is indistinguishable from MFJP — so reputation contributes nothing and calibration actively costs. Two measured properties explain it: the production judges differ in bias but barely in *scale* (sd ratio 1.19), and their errors are strongly correlated ($\bar{\rho} = 0.729$) despite spanning three model families. In simulation the same picture holds — under the panel’s own assumptions a plain mean matches MFJP to 0.0004 Spearman and cost-matched self-consistency beats it — and we identify scale heterogeneity as the *necessary condition* for the machinery to pay: when we manufacture it, the ladder is clean ($0.938 \rightarrow 0.948 \rightarrow 0.958$), but a batch Gaussian EM estimator and the best single judge at one third the cost both still beat MFJP. At the correlation production actually exhibits, a single judge at one third the cost is statistically indistinguishable from the panel ($\Delta\text{AUC} -0.004$, $[-0.055, +0.045]$). Reputation is actively *harmful* when the majority is systematically biased: it down-weights the one accurate dissenter to 0.26 while the two confounded judges keep 0.41–0.46, costing 0.039 Spearman versus calibration alone. The panel’s one clear, large win is adversarial: when responses are engineered to game a single judge, MFJP scores 0.918 against 0.553 for cost-matched self-consistency — diversity, not the mean-field machinery, is what buys this. We also report that the claimed “free confidence signal” is uninformative in benign simulation ($\rho = 0.036$) and *negative* under distribution drift, though it does work on real data (+0.331); and that z-scoring destroys absolute calibration: at a realistic 0.9 base rate MFJP attains 0.657 accuracy and 0.456 ECE where a plain mean attains 0.979 and 0.190. Finally, three defects in the shipped code: the live config names a retired model so the “3-judge” panel silently runs as two; the process-global mutex is held across all sequential judge HTTP calls, serialising judged traffic for up to 90s; and calibration state is in-process only, so every deploy re-enters cold start.

1 The question

A single LLM judge has one model’s bias, one point of failure, and one thing to game. The shipped answer at Hanzo is the Mean-Field Judge Panel, live as [claude-3-5-haiku, gpt-4o-mini, llama-3.1-8b] at sample 0.10. It does two things beyond averaging a committee:

1. **Calibration.** Each judge’s raw score is z-scored against that judge’s own running mean and standard deviation (Welford, online) and logistic-squashed, so a harsh judge’s 0.6 and a lenient judge’s 0.8 both read as “above my own bar.”
2. **Reputation.** Each judge carries a reliability weight updated multiplicatively, $w \leftarrow (1 - \pi) w e^{-\lambda|c-r|} + \pi$, with $\lambda = 2.0$, $\pi = 0.1$, $w_{\min} = 0.05$. A judge that tracks the calibrated consensus keeps its weight; one that drifts is exponentially suppressed.

The reward is the reliability-weighted calibrated consensus; panel disagreement is returned as a free confidence signal. The design is computationally cheap by construction — four floats per judge, $O(1)$ per call.

The question this paper answers is not “does a panel beat one judge” (it usually does, and a companion paper measured that). It is narrower and more consequential: **does the mean-field machinery earn its keep over simply averaging the same diverse judges?** And at equal budget — three judges cost three times one judge — does the panel beat what else that budget could buy?

1.1 What the prior work established, and what it did not

A companion Hanzo study measured judge panels against executable ground truth and found that a panel buys *variance*, not accuracy: a five-judge majority reached 0.750 against a mean single-judge 0.685, but the best single rubric alone reached 0.787. It also reported a negative: a mean-field construction in which judges *see* the aggregate and revise *herds*, scoring below plain majority.

Two things leave our question open. First, that study’s five judges shared one model and differed only by rubric, so it measured rubric diversity, not model diversity — a limitation it flagged itself. Second, and more importantly, its “mean-field” arm is a *different algorithm* from the shipped MFJP: it exposes the consensus to the judges, whereas MFJP aggregates independent scores post hoc and never lets judges see each other. The herding result therefore says nothing about the shipped code. Nobody has ablated it.

1.2 The shipped test begs the question

The repository already contains a Monte-Carlo test, `TestMFJPRecoversGroundTruth`. It is confirmatory in three ways, and we flag this because it is the existing evidence base. Its judges have *independent* Gaussian errors around the truth — which is precisely the panel’s own generative assumption. Its single adversary emits *pure noise*, the easiest possible case for reputation to suppress. Its metric is Spearman rank correlation, which is invariant to exactly the per-judge additive bias that calibration removes, so calibration cannot be penalised. It then *asserts* that MFJP wins, comparing only against a naive mean that conflates calibration and reputation into one arm. We do not assert. We measure, and we choose conditions that can falsify.

2 Method

2.1 Provenance

All results drive the shipped algorithm. `router_judge_panel.go` was copied verbatim from the published module `github.com/hanzoai/ai@v1.826.4` — checksum-verified from the module zip, byte-identical below the `package` line — into a standalone harness whose only addition is a shim supplying the two external symbols the file references. The algorithm is untouched; `panelScore` is the production function.

2.2 Track A — real judges, executed ground truth

We take 164 HumanEval problems and the candidate program generated for each by Qwen2.5-Coder-7B, with pass/fail labels obtained by *executing* the evalplus test suite (89 pass, 75 fail; base rate 0.543, so a constant-PASS predictor scores 0.543). Three judges from genuinely different model families score each candidate using the *verbatim* production rubric from judgeRubric — same system prompt, same strict JSON contract, `temperature 0`, `max_tokens 256`. No judge sees the tests or the labels. This closes the model-diversity gap the prior study flagged.

One substitution was forced on us and is itself a finding (§4): the live config names `claude-3-5-haiku`, which the gateway no longer serves. We substituted its successor `claude-haiku-4.5`.

2.3 Track B — simulation that can falsify

Track A gives one point in the space. Only simulation can *sweep* the inter-judge error correlation $\bar{\rho}$ and locate the boundary where the panel stops paying. We generate

$$\text{raw}_{ij} = \text{clip}_{[0,1]}(q_i + b_j + g_j c_i + \varepsilon_{ij}), \quad \varepsilon_{ij} \sim \mathcal{N}(0, s_j^2), \quad c_i \sim \mathcal{N}(0, 1),$$

where q_i is latent quality, b_j is judge j ’s bias, and c_i is a *shared confound* — a property such as verbosity or fluency that judges reward but that is orthogonal to quality. The error $e_{ij} = b_j + g_j c_i + \varepsilon_{ij}$ is independent across judges only when $g = 0$. The pairwise error correlation is $\rho_{jk} = g_j g_k / \sqrt{(g_j^2 + s_j^2)(g_k^2 + s_k^2)}$, so g is the dial that moves us from the panel’s assumed world to the one real LLM judges inhabit, where shared pretraining and shared RLHF make errors correlate. Dawid–Skene assumes conditional independence too, so it should degrade alongside. We report the *realized* $\bar{\rho}$ measured from the generated data, not the nominal target, because clipping attenuates it.

400 independent replicates of 600 items each. Online arms are scored on the post-warmup half so that calibration and reputation have converged — a choice that *favours* MFJP; cold start is reported separately.

2.4 The surgical ablation

This is the core of the design. Three arms differ by exactly one mechanism each:

- **raw mean** — unweighted mean of raw scores. No calibration, no reputation.
- **z-mean** — *identical* online calibration to MFJP: the same `calibrateScore` function, the same `judgeStat` Welford state, the same score-then-observe ordering. Reputation weights pinned to 1.
- MFJP — the shipped `panelScore`.

Any MFJP–z-mean gap is attributable to *reputation alone*; any z-mean–raw-mean gap to *calibration alone*. The shipped test conflates both into a single “naive mean” comparison and so cannot attribute credit.

2.5 Competing arms

Beyond the ablation ladder: **majority vote** (each judge votes against its own running mean); **Dawid–Skene**, the classical latent-confusion-matrix EM aggregator, run in *batch* over all items — a deliberate advantage over the online arms; **Gaussian EM**, the principled continuous competitor, which is the maximum-likelihood estimator under *exactly the panel’s own assumed model* (per-judge

additive bias plus per-judge Gaussian noise, conditionally independent) and therefore performs bias removal plus inverse-variance weighting — MFJP is best understood as a cheap online heuristic approximating this batch optimum; **cascade** (cheap judge first, escalate to the panel only near its decision boundary); and two **cost-matched** arms that spend the panel’s three-call budget on one judge — **self-consistency** (one judge sampled three times, averaged) and a **homogeneous panel** (three samples of one judge fed to MFJP).

2.6 Metrics

Spearman correlation with latent quality (ranking); AUC; Brier score and expected calibration error over 10 bins (the confidence claim); accuracy; cost in judge calls. For the confidence signal we correlate MFJP’s returned confidence against *rank* error — $|\text{rank}(\hat{q}) - \text{rank}(q)|$ — because MFJP’s z-squashed output is not on the same scale as q and a raw $|\hat{q} - q|$ would be scale-confounded and unfair to it. Differences are reported as *paired* deltas versus MFJP with 95% bootstrap confidence intervals (4000 resamples over replicates; 5000 over items for Track A).

3 Results

3.1 Track A — real judges on executed ground truth

Three real judges scored 164 execution-labelled candidates. 44 of 492 calls (8.9%) failed to yield a parseable score even after retries and are recorded as abstentions. All arms — including MFJP — were run on the same matrix with abstentions imputed at the judge’s own mean, so no arm sees more information than another.

judge	mean	sd	AUC	best acc	abstentions
claude-haiku-4.5	0.487	0.323	0.800	0.744	18
gpt-4o-mini	0.734	0.343	0.821	0.805	16
llama-3.1-8b	0.763	0.386	0.750	0.750	10

Table 1: Per-judge behaviour on 164 items, base rate 0.543.

Two measurements decide how the rest of this paper applies to production.

Finding 1. The production judges differ in bias but not in scale. Their means span 0.487 to 0.763 — a large 0.28 offset gap, with Haiku much harsher — but their standard deviations are 0.323, 0.343, 0.386, a max/min ratio of only **1.19**. R8 established that scale heterogeneity is the *necessary condition* for calibration and reputation to pay. It is absent here. The mean-field machinery is deployed in precisely the regime where the simulation predicts it is inert.

Finding 2. The real judges are heavily correlated: $\bar{\rho} = 0.729$, with pairwise score correlations of 0.575, 0.533 and 0.655. Model-family diversity does *not* deliver error independence — shared pretraining data and shared preference tuning dominate. Reading $\bar{\rho} = 0.73$ off the R2 sweep places production near the far end of the curve, where the diversity premium has decayed from 0.044 to 0.026.

arm	AUC	best acc	acc@0.5	Brier	ECE	cost
MFJP (calib+reputation)	0.824	0.774	0.738	0.183	0.135	3.00
z-mean diverse (calib only)	0.827	0.780	0.738	0.181	0.131	3.00
raw mean diverse	0.859	0.817	0.750	0.166	0.121	3.00
Gaussian EM (batch)	0.863	0.823	0.799	0.177	0.140	3.00
Dawid–Skene (batch EM)	0.736	0.732	0.457	0.542	0.542	3.00
majority vote	0.821	0.750	0.750	0.175	0.096	3.00
cascade	0.806	0.756	0.738	0.187	0.132	1.50
single: claude-haiku-4.5	0.800	0.744	0.732	0.190	0.123	1.00
single: gpt-4o-mini	0.821	0.805	0.652	0.201	0.204	1.00
single: llama-3.1-8b	0.750	0.750	0.701	0.257	0.251	1.00

Table 2: Real judges, executed ground truth, $n = 164$, $\bar{\rho} = 0.729$.

Paired bootstrap over items (5000 resamples), Δ AUC versus MFJP: raw mean +0.0348 [+0.0142, +0.0567]*; Gaussian EM +0.0381 [+0.0072, +0.0699]*; z-mean +0.0022 [−0.0054, +0.0099]; majority −0.0039 [−0.0323, +0.0240]; **gpt-4o-mini** alone −0.0039 [−0.0551, +0.0447]; **claude-haiku-4.5** alone −0.0247 [−0.0693, +0.0207]; Dawid–Skene −0.0884 [−0.1401, −0.0356]*.

Finding 3. On real judges and real executed ground truth, a plain mean of the same three judges significantly outperforms the shipped panel: AUC 0.859 versus 0.824, $\Delta = +0.0348$ with the interval excluding zero. The z-mean ablation is statistically indistinguishable from MFJP (+0.0022), so reputation contributes nothing; the raw-mean-versus-z-mean gap shows the loss is attributable to *calibration*, which — consistent with Finding 1 — has no scale heterogeneity to correct and merely discards information.

Finding 4. At $\bar{\rho} = 0.73$, a single judge at one third the cost is statistically indistinguishable from the panel. **gpt-4o-mini** alone scores AUC 0.821 against the panel’s 0.824 ($\Delta = -0.0039$, [−0.0551, +0.0447]), and on best-threshold accuracy it is higher (0.805 vs 0.774). This is the cost-matched boundary the simulation predicted: the diversity premium does not survive the error correlation that real judges actually exhibit.

Robustness. Dropping every item on which any judge abstained leaves 122 complete cases and no imputation anywhere. The conclusions are unchanged: raw mean +0.0280 [+0.0072, +0.0503]*, Gaussian EM +0.0415 [+0.0104, +0.0767]*, z-mean +0.0016 [−0.0074, +0.0110] versus MFJP. The result is not an artifact of how abstentions were handled.

The confidence signal does work here. On real data MFJP’s disagreement signal correlates +0.331 with correctness: mean absolute error is 0.344 on the high-confidence half against 0.448 on the low-confidence half. This is the one component that survives Track A intact, and it is consistent with the simulation — $\bar{\rho} = 0.73$ with a 0.28 spread in judge means is exactly the structural disagreement the signal is able to detect.

Dawid–Skene fails on real data (AUC 0.736, Brier 0.542). Binarising a continuous judge score at each judge’s median discards the magnitude information that carries most of the signal here. The classical aggregator is the wrong tool when judges emit graded scores rather than categorical labels — a point worth recording, since it is the aggregator the literature would nominate.

3.2 R1 — the panel’s own assumptions

Independent errors, homogeneous scales, realized $\bar{\rho} = 0.047$: the regime the design assumes and the shipped test simulates.

arm	Spearman	AUC	Brier	ECE	acc	cost
MFJP (calib+reputation)	0.970	0.990	0.112	0.258	0.943	3.00
z-mean diverse (calib only)	0.971	0.990	0.112	0.258	0.943	3.00
raw mean diverse	0.971	0.990	0.092	0.209	0.943	3.00
Gaussian EM (batch)	0.933	0.972	0.099	0.163	0.906	3.00
Dawid–Skene (batch EM)	0.901	0.976	0.059	0.051	0.934	3.00
majority vote	0.903	0.976	0.052	0.030	0.935	3.00
cascade	0.947	0.987	0.106	0.243	0.943	1.74
self-consistency ($\times 3$)	0.973	0.991	0.088	0.197	0.945	3.00
homogeneous panel ($\times 3$)	0.972	0.990	0.113	0.260	0.944	3.00
single judge (best, oracle)	0.932	0.973	0.115	0.172	0.846	1.00
single judge (random draw)	0.926	0.972	0.121	0.176	0.829	1.00

Table 3: R1, independent errors, $\bar{\rho} = 0.047$, 400 replicates.

Finding 5. Under its own assumptions the mean-field machinery adds **nothing**. Paired deltas versus MFJP: z-mean +0.0002 [+0.0001, +0.0002], raw mean +0.0004 [+0.0004, +0.0005]. Both intervals exclude zero — so the plain mean is *significantly better* — but the effect size is four ten-thousandths of a correlation unit. The honest statement is that they are indistinguishable in practice, and that neither calibration nor reputation buys anything here.

The reason is structural rather than empirical. Constant per-judge bias shifts every item equally, so it cannot affect *ranking*; Spearman is invariant to it. Calibration’s stated purpose — removing each judge’s standard — is therefore irrelevant to rank quality when judges differ only in offset. That leaves absolute level, which z-scoring actively destroys (§3.6).

Finding 6. At equal cost the panel loses. Self-consistency — one judge sampled three times — scores 0.973 versus 0.970 (+0.0023, [+0.0020, +0.0026]). Three calls spent on one judge beat three calls spread across three judges in benign conditions.

3.3 R2 — sweeping $\bar{\rho}$: where the premium goes

$\bar{\rho}$	MFJP	z-mean	Gaussian EM	1 judge	self-cons.
0.047	0.970	0.971	0.933	0.926	0.973
0.124	0.962	0.963	0.925	0.919	0.965
0.203	0.953	0.953	0.916	0.911	0.955
0.284	0.941	0.941	0.904	0.900	0.943
0.367	0.926	0.926	0.890	0.887	0.927
0.452	0.906	0.906	0.871	0.868	0.907
0.540	0.877	0.878	0.845	0.842	0.878
0.632	0.835	0.835	0.806	0.803	0.836
0.728	0.764	0.764	0.740	0.738	0.764
0.831	0.621	0.621	0.607	0.602	0.621

Table 4: Spearman versus realized inter-judge error correlation. 400 replicates per row.

Two readings. First, the **diversity premium** — MFJP minus a randomly drawn single judge — decays monotonically from 0.044 at $\bar{\rho} = 0.05$ to 0.019 at $\bar{\rho} = 0.83$: as judges correlate, the third judge stops adding information, exactly as theory predicts. Second, and more damaging, MFJP and z-mean are **identical to three decimal places at every $\bar{\rho}$** , and cost-matched self-consistency equals or beats MFJP at every $\bar{\rho}$. The reputation mechanism contributes nothing anywhere along this sweep.

3.4 R3 — reputation is harmful when the majority is biased

Two judges load on the shared confound ($g = 0.35$); one is accurate ($g = 0$). Realized $\bar{\rho} = 0.315$.

arm	Spearman	AUC	Brier	ECE	acc	cost
MFJP	0.768	0.890	0.155	0.131	0.794	3.00
z-mean (calib only)	0.807	0.910	0.149	0.151	0.814	3.00
raw mean	0.790	0.900	0.132	0.069	0.809	3.00
Gaussian EM (batch)	0.629	0.818	0.182	0.096	0.737	3.00
Dawid–Skene (batch EM)	0.727	0.875	0.220	0.222	0.775	3.00
majority vote	0.756	0.891	0.134	0.111	0.777	3.00
self-consistency ($\times 3$)	0.632	0.820	0.178	0.088	0.742	3.00
single judge (best)	0.928	0.972	0.096	0.149	0.904	1.00
single judge (random)	0.720	0.865	0.155	0.116	0.791	1.00

Table 5: R3, biased majority with one accurate dissenter, $\bar{\rho} = 0.315$, 400 replicates.

Finding 7. Reputation **actively hurts**: MFJP 0.768 against z-mean 0.807, a loss of 0.039 attributable to the reputation mechanism alone. The mechanism is visible in the weights. After 600 items the accurate dissenter holds weight **0.260** while the two confounded judges hold **0.412** and **0.460**. Reputation cannot distinguish “drifting from the consensus because it is wrong” from “drifting from the consensus because the consensus is wrong,” so it suppresses the only judge telling the truth by a factor of 1.7.

Note also that the best single judge — the honest one — scores 0.928 at *one third* the cost, beating every three-judge arm. When a shared bias exists and one judge is free of it, aggregation is the wrong move entirely; judge selection is the right one.

3.5 R4 — adversarial gaming: the panel’s real win

12% of items are engineered to spike one judge while true quality is low.

arm	Spearman	AUC	acc	cost
MFJP	0.918	0.990	0.946	3.00
z-mean (calib only)	0.910	0.989	0.945	3.00
raw mean	0.902	0.987	0.944	3.00
Gaussian EM (batch)	0.928	0.977	0.915	3.00
Dawid–Skene (batch EM)	0.868	0.979	0.930	3.00
majority vote	0.841	0.974	0.941	3.00
self-consistency (×3)	0.553	0.797	0.833	3.00
homogeneous panel (×3)	0.551	0.796	0.819	3.00
single judge (random)	0.793	0.914	0.809	1.00

Table 6: R4, reward hacking against one judge, $\bar{\rho} = 0.101$, 400 replicates.

Finding 8. This is the panel’s genuine, large win, and the only regime where reputation clearly pays. MFJP 0.918 versus cost-matched self-consistency 0.553 — a gap of 0.365 $[-0.371, -0.360]$. Sampling one judge three times is fooled every time, because it is the same judge with the same exploitable bias. Reputation also earns its keep here: MFJP beats z-mean by +0.0089 and raw mean by +0.0168, both intervals excluding zero.

The credit belongs mostly to *diversity*, not to the mean-field machinery: raw mean over diverse judges already reaches 0.902 of the 0.918. But the machinery does add a real increment here, and this is the argument that justifies the panel’s existence.

3.6 R5 — z-scoring destroys absolute calibration

Quality $\sim$ Beta(5, 1): most responses are genuinely good, which is the regime a production serving path actually occupies.

arm	Spearman	AUC	Brier	ECE	acc
MFJP	0.830	0.991	0.240	0.456	0.657
z-mean (calib only)	0.831	0.991	0.240	0.456	0.655
raw mean	0.828	0.992	0.056	0.190	0.979
Gaussian EM (batch)	0.762	0.972	0.055	0.184	0.970
Dawid–Skene (batch EM)	0.708	0.840	0.464	0.571	0.371
self-consistency	0.845	0.992	0.048	0.159	0.981
single judge (random)	0.691	0.972	0.080	0.182	0.907

Table 7: R5, base rate ≈ 0.9 , $\bar{\rho} = 0.031$, 400 replicates.

Finding 9. MFJP’s reward is **purely relative** and its absolute calibration is catastrophic where it matters most. Because `calibrateScore` z-scores against the judge’s own running mean and

logistic-squashes, the output is forced toward a spread around 0.5 regardless of how good the responses actually are. When 90% of responses are good, MFJP attains accuracy 0.657 and ECE 0.456; a plain mean of the same judges attains 0.979 and 0.190. Note that AUC is *identical* (0.991 vs 0.992) — ranking is preserved, absolute level is destroyed. A reward consumer that thresholds the score, rather than only ranking by it, is being misled on roughly one third of good responses.

This is not a tuning problem; it is what z-scoring means. Any downstream use of the MFJP scalar as a probability or as an absolute quality gate is unsound by construction.

3.7 R8 — the one regime where the machinery earns its keep

To avoid rigging the ablation, we constructed the case calibration is *for*: judges that differ in *spread*, not merely offset (noise $s = 0.30, 0.05, 0.12$). A raw mean is dominated by whichever judge swings widest; calibration equalises influence.

arm	Spearman	AUC	Brier	ECE	cost
MFJP	0.958	0.987	0.117	0.254	3.00
z-mean (calib only)	0.948	0.983	0.120	0.246	3.00
raw mean	0.938	0.977	0.100	0.183	3.00
Gaussian EM (batch)	0.983	0.995	0.089	0.223	3.00
Dawid–Skene (batch EM)	0.883	0.982	0.045	0.041	3.00
self-consistency ($\times 3$)	0.860	0.940	0.117	0.130	3.00
single judge (best)	0.983	0.995	0.116	0.228	1.00

Table 8: R8, heterogeneous judge scales, $\bar{\rho} = 0.089$, 400 replicates.

Finding 10. Both mechanisms do earn their keep here, and the ladder is clean and monotone: raw mean 0.938 $\rightarrow$ z-mean 0.948 (calibration, +0.0099) $\rightarrow$ MFJP 0.958 (reputation, +0.0099), both intervals excluding zero. **Scale heterogeneity is the necessary condition for the mean-field machinery to pay.** But even here MFJP is beaten by Gaussian EM (0.983, +0.0259) and matched by the best single judge at one third the cost.

This makes the design decision empirical rather than theoretical: the machinery pays only if the deployed judges actually differ in scale. §3.1 measures whether they do.

3.8 The confidence signal

Panel disagreement is advertised as a free confidence signal. Correlating it against rank error across regimes:

regime	$\rho(\text{confidence}, - \text{rank error})$
R1 independent errors	+0.036
R3 biased majority	+0.561
R4 reward hacking	+0.256
R5 high base rate	+0.034
R6 non-stationary quality	−0.143
R8 heterogeneous scales	+0.271

Table 9: Informativeness of the returned confidence.

Finding 11. The confidence signal is **uninformative in benign conditions** (+0.036) and **actively misleading under drift** (−0.143: the panel is most confident when it is most wrong). It is genuinely informative only when there is structural disagreement to detect — a biased faction (+0.561), an adversarial item (+0.256), or scale heterogeneity (+0.271). It should be used as a *disagreement detector*, which is what it measures, and never as a general-purpose reliability estimate.

3.9 Cold start

Scored on the first 8% of the stream with no warmup, MFJP falls to 0.965 while the stateless raw mean holds 0.971 and self-consistency 0.973. The online state is a small liability before it converges, which matters more than it appears (§4).

4 Three defects in the shipped code

These are independent of the statistics and were found by reading and exercising the shipped artifact.

1. The panel silently runs with two judges, not three. The live configuration names `claude-3-5-haiku`. That model was retired on 2026-02-19 and the gateway returns HTTP 400 for it. In `panelScore` a judge whose call fails is skipped by `continue` — “an erroring judge abstains; the panel stays resilient.” The resilience is real, and it is precisely what conceals the fault: the advertised three-judge mean-field panel has been running as a two-judge panel with no error surfaced. Judge abstentions need a metric.

2. A process-global mutex is held across all judge network calls. `panelScore` takes `panelState.mu.Lock()` and defers the unlock *before* the loop that issues the judge HTTP requests. The judges are called **sequentially** — there are no goroutines in the function — and `judgeClient` carries a 30 s timeout. So a single judged turn can hold a process-wide lock for up to $3 \times 30 = 90$ s, during which every other judged turn in the process blocks. Panel latency is additive ($\sum$, not $\max$) and the panel is a global serialisation point. The fix is to issue the calls concurrently outside the lock and take it only to fold results into state.

3. Calibration state is in-process and unpersisted, so every deploy resets it. `panelState` is a package-level variable with no persistence. Every pod restart returns all judges to $n=0$, where `std()` falls back to a 0.2 prior and the first score is z-scored against a mean of zero — a raw 0.7 calibrates to 0.97. At the production **sample 0.10** rate, only one turn in ten contributes state, so the panel spends a long window after every rollout in the cold-start regime of §3.8, emitting inflated rewards.

5 Recommendation

The evidence supports one recommendation, in three parts.

Keep the diverse panel; drop the mean-field machinery. Replace the reliability-weighted calibrated consensus with a plain mean of raw diverse judge scores. On real judges against executed ground truth it beats MFJP by 0.035 AUC with the interval excluding zero; in simulation it matches MFJP to 0.0004 Spearman under benign conditions, beats it by 0.039 when the majority is biased,

and is dramatically better calibrated in absolute terms (0.056 vs 0.240 Brier, 0.979 vs 0.657 accuracy at a realistic base rate). The necessary condition for the machinery to pay — judges differing in scale — is measurably absent from the deployed panel (sd ratio 1.19). Diversity is doing the work that matters — the entire adversarial robustness result survives, since raw mean over diverse judges reaches 0.902 of MFJP’s 0.918 in R4. Calibration and reputation are worth reinstating only if the deployed judges are measured to differ in *scale*, and reputation only if adversarial gaming is an active threat; if it is, weight by *held-out agreement with ground truth*, never by agreement with the current consensus, which is what makes R3 fail.

Do not use the score as a probability. The z-scored output is a ranking signal. Any consumer thresholding it as an absolute quality gate should consume a raw mean instead.

Fix the three defects in §4 regardless of the aggregation choice; the concurrency one is a live production risk independent of this study.

If the goal is peak accuracy rather than architectural simplicity, the batch Gaussian EM estimator dominated MFJP in every regime where the machinery mattered at all (+0.026 in R8) and is the principled version of what MFJP approximates — MFJP’s defensible claim was never accuracy but that it is an $O(1)$ -state streaming approximation to it. That claim is true; the measurements show the approximation is not buying enough over a plain mean to justify its failure modes.

6 Threats to validity

Track B is *simulation* and its conclusions are conditional on the generative model. We chose that model to be able to falsify the panel — correlated errors, a systematically biased majority, non-stationarity — rather than to reproduce its assumptions, and we report the regime (R8) where the machinery wins. But a real judge population may be structured in ways this parameterisation does not capture; the R2 sweep is a statement about a one-parameter confound family, not about all forms of dependence. Latent quality is a scalar, which understates multi-dimensional disagreement.

Track A is $n = 164$ single-seed with three judges in one domain (code) where an oracle exists; the extrapolation to non-verifiable work is exactly the extrapolation the method cannot test. Its candidates come from one generator, so failure modes are not representative of all model errors. Substituting `claude-haiku-4.5` for the retired `claude-3-5-haiku` means Track A measures the panel as *intended*, not the panel as *currently running*.

Bootstrap intervals quantify sampling variability only; they do not cover model misspecification, which is the dominant uncertainty in Track B. Finally, several deltas here are statistically significant and practically negligible — that is a property of 400 replicates, and we have tried throughout to report effect sizes rather than lean on interval exclusion.

7 Conclusion

The Mean-Field Judge Panel’s population of diverse judges is worth keeping: it is what defeats reward hacking, and by a large margin. The mean-field machinery layered on top of that population is not. On real judges against executed ground truth, plainly averaging the same three scores beats the shipped panel by 0.035 AUC; the calibration-only ablation is indistinguishable from the full mechanism, so reputation contributes nothing and calibration costs. In simulation the mechanisms are worth four ten-thousandths of a correlation unit, in the wrong direction, and are inert across the whole $\bar{\rho}$ sweep; reputation is affirmatively harmful when the majority shares a bias, for the structural reason that agreement with the consensus is not evidence when the consensus is what is in error; and the calibration step destroys the absolute scale that a thresholding reward consumer

depends on. The machinery earns its keep in exactly one measured condition — judges that differ in scale — and that condition is measurably absent from the deployed panel, whose judges differ in bias but share a scale to within a factor of 1.19 and whose errors correlate at $\bar{\rho} = 0.729$ despite three distinct model families. Diverse judges, plainly averaged, is the configuration the evidence supports.

Reproduction

The harness drives the shipped `panelScore` copied verbatim from the checksum-verified module github.com/hanzoai/ai@v1.826.4. It contains the generative regimes, the surgical ablation, all competing arms, the paired bootstrap, and the Track A collector that records the per-judge per-item score matrix — an artifact the prior instrument computed in memory and discarded.